Mini-Project 1B 4001

Shokhsanam Uktamova and Caroline Lewis

Experiment 1 (run on local)

1. **Get familiar with what EEG data are, electrode positioning systems and the notion of a montage. Summarize the advantages and disadvantages of EEG recordings and differences between bipolar and referential montages.**

Advantages and Disadvantages of EEG Recordings

Advantages	Disadvantages
• High temporal resolution • Non-invasive • Relatively inexpensive • Portable options • Useful for dynamic brain activity	• Poor spatial resolution • Highly susceptive to artifacts • Determining the exact, unique neuronal generators of the recorded signals is very challenging

Differences between Bipolar and Referential Montages

	Bipolar Montage	**Referential Montage**
Def.	It measures voltage difference between two adjacent electrodes	Measures voltage of each electrode relative to a common reference electrode
Focus	Highlights local differences between neighboring regions	Shows absolute activity relative to a fixed point
Pros	Reduces common noises, easier to see localized events	Can see overall activity, good for comparing across regions
Cons	Can obscure global patterns, harder to compare across subjects	Noise/artifacts in the reference affect all channels, less sensitive to local contrasts
Usage	Clinical seizure localization, detecting local patterns	Research studies, general mapping of brain rhythms

2. **Read about the dataset structure. What is the total recording time for each participant? You are going to work with the data from participant S01 placed in the data/ folder.**
   - The total recording time for each participant is 20 minutes. They all played 4 games for 5 minutes (4x5=20).

3. **Open the eeg_emotion_classification.ipynb notebook. If it asks you to install additional modules, kernel, etc., choose "Install". Follow the notebook to process the dataset and train a model. You can adjust the parameters specified by "TODO" comments to increase accuracy.**
   - Done

4. **Review the Heart project to learn how to display images on the micro:bit. Choose an image for each emotion and write a script for projecting that image to the micro:bit (see flicker.py and check_output.py from the previous project for reference). Integrate the projection functionality into the notebook so that when the network makes a prediction, it gets displayed on the micro:bit.**

Experiment 2 (run on remote)

Transform the notebook from Experiment 1 into a script that can run on a terminal-first machine (e.g., FABRIC). Train the network on a remote machine and compare the training time between local and remote.

Local Terminal Run: Training time was 21.805 seconds.

```
~/Documents/CI-BioEng-Class/emotion_recognition  main ● ?
$ python eeg_emotion_classification_2.py
Epoch 75: train_loss=1.3038, train_acc=0.3678, test_loss=1.3455, test_acc=0.3265
Epoch 76: train_loss=1.3013, train_acc=0.3662, test_loss=1.3431, test_acc=0.3360
Epoch 77: train_loss=1.2992, train_acc=0.3715, test_loss=1.3407, test_acc=0.3360
Epoch 78: train_loss=1.2961, train_acc=0.3739, test_loss=1.3381, test_acc=0.3319
Epoch 79: train_loss=1.2919, train_acc=0.3793, test_loss=1.3356, test_acc=0.3387
Epoch 80: train_loss=1.2917, train_acc=0.3779, test_loss=1.3330, test_acc=0.3428
Epoch 81: train_loss=1.2891, train_acc=0.3952, test_loss=1.3305, test_acc=0.3482
Epoch 82: train_loss=1.2883, train_acc=0.3945, test_loss=1.3279, test_acc=0.3523
Epoch 83: train_loss=1.2830, train_acc=0.4025, test_loss=1.3250, test_acc=0.3550
Epoch 84: train_loss=1.2832, train_acc=0.3995, test_loss=1.3227, test_acc=0.3564
Epoch 85: train_loss=1.2809, train_acc=0.4048, test_loss=1.3196, test_acc=0.3564
Epoch 86: train_loss=1.2743, train_acc=0.4102, test_loss=1.3169, test_acc=0.3564
Epoch 87: train_loss=1.2723, train_acc=0.4108, test_loss=1.3142, test_acc=0.3550
Epoch 88: train_loss=1.2673, train_acc=0.4168, test_loss=1.3115, test_acc=0.3550
Epoch 89: train_loss=1.2651, train_acc=0.4134, test_loss=1.3087, test_acc=0.3591
Epoch 90: train_loss=1.2650, train_acc=0.4105, test_loss=1.3063, test_acc=0.3605
Epoch 91: train_loss=1.2591, train_acc=0.4227, test_loss=1.3031, test_acc=0.3632
Epoch 92: train_loss=1.2566, train_acc=0.4181, test_loss=1.3006, test_acc=0.3632
Epoch 93: train_loss=1.2554, train_acc=0.4220, test_loss=1.2976, test_acc=0.3673
Epoch 94: train_loss=1.2524, train_acc=0.4225, test_loss=1.2945, test_acc=0.3727
Epoch 95: train_loss=1.2481, train_acc=0.4217, test_loss=1.2920, test_acc=0.3741
Epoch 96: train_loss=1.2443, train_acc=0.4217, test_loss=1.2899, test_acc=0.3754
Epoch 97: train_loss=1.2440, train_acc=0.4278, test_loss=1.2866, test_acc=0.3768
Epoch 98: train_loss=1.2380, train_acc=0.4300, test_loss=1.2840, test_acc=0.3822
Epoch 99: train_loss=1.2356, train_acc=0.4365, test_loss=1.2818, test_acc=0.3836
Training time (seconds): 21.805314779281616
Predicted emotion: horror
Predicted emotion (micro:bit skipped): horror

~/Documents/CI-BioEng-Class/emotion_recognition  main ● ?
$
```

Fabric Remote Terminal Run: Training time was 10.85 seconds.

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```
ubuntu@testnode: ~/CI-BioEng-Class/emotion_recognition
Epoch 65: train_loss=1.3245, train_acc=0.3550, test_loss=1.3669, test_acc=0.3034
Epoch 66: train_loss=1.3195, train_acc=0.3628, test_loss=1.3649, test_acc=0.3048
Epoch 67: train_loss=1.3214, train_acc=0.3592, test_loss=1.3632, test_acc=0.3007
Epoch 68: train_loss=1.3192, train_acc=0.3675, test_loss=1.3610, test_acc=0.3061
Epoch 69: train_loss=1.3172, train_acc=0.3675, test_loss=1.3588, test_acc=0.3116
Epoch 70: train_loss=1.3122, train_acc=0.3642, test_loss=1.3573, test_acc=0.3061
Epoch 71: train_loss=1.3131, train_acc=0.3627, test_loss=1.3550, test_acc=0.3129
Epoch 72: train_loss=1.3115, train_acc=0.3645, test_loss=1.3526, test_acc=0.3231
Epoch 73: train_loss=1.3079, train_acc=0.3683, test_loss=1.3501, test_acc=0.3271
Epoch 74: train_loss=1.3065, train_acc=0.3666, test_loss=1.3481, test_acc=0.3285
Epoch 75: train_loss=1.3038, train_acc=0.3678, test_loss=1.3455, test_acc=0.3265
Epoch 76: train_loss=1.3013, train_acc=0.3662, test_loss=1.3431, test_acc=0.3360
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Epoch 97: train_loss=1.2440, train_acc=0.4278, test_loss=1.2866, test_acc=0.3768
Epoch 98: train_loss=1.2380, train_acc=0.4300, test_loss=1.2840, test_acc=0.3822
Epoch 99: train_loss=1.2356, train_acc=0.4365, test_loss=1.2818, test_acc=0.3836
Training time (seconds): 10.84970998764038
Predicted emotion: horror
Predicted emotion (micro:bit skipped): horror
(emotion-recognition) ubuntu@testnode:~/CI-BioEng-Class/emotion_recognition$
```

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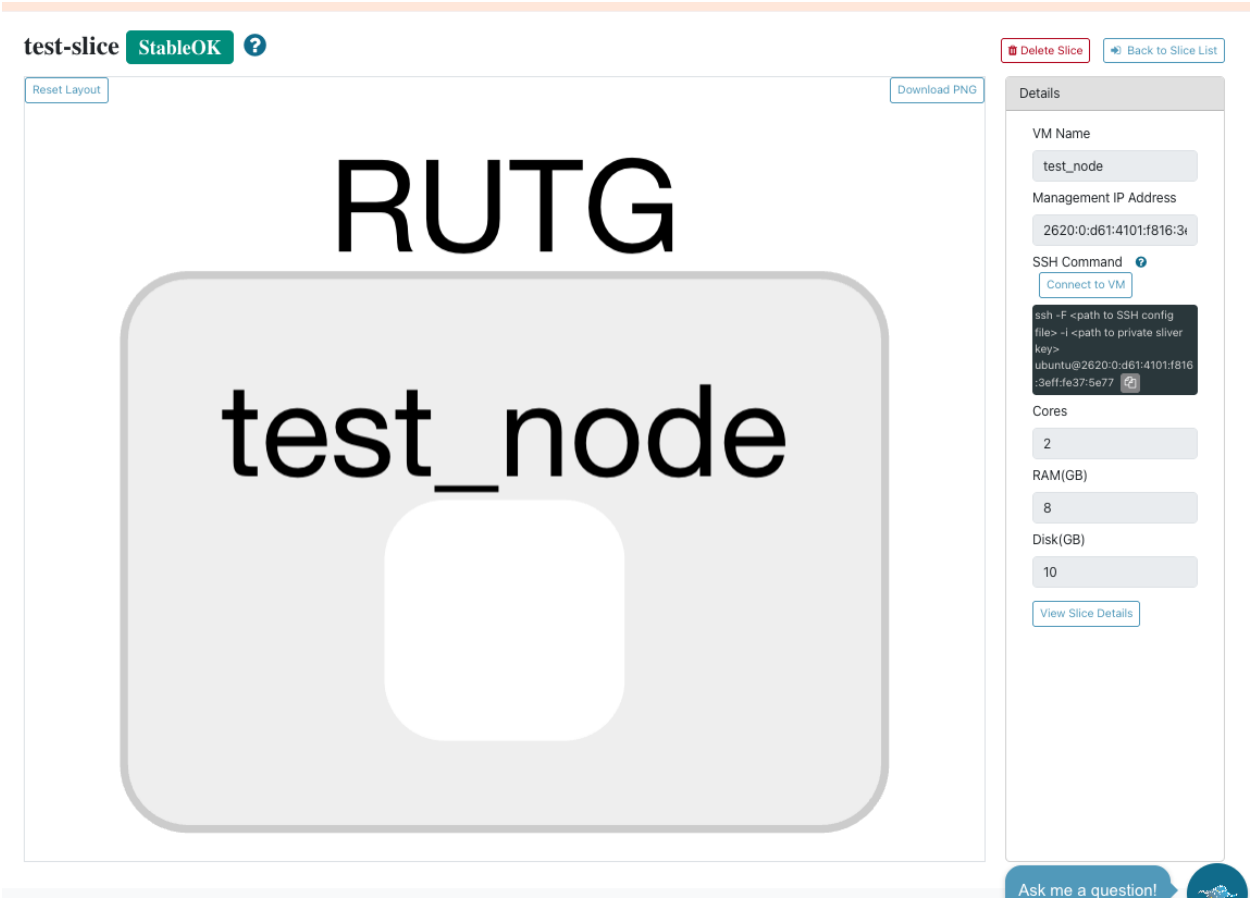

To submit

1. **A Word document with answers to questions 1-2 from Experiment 1 and with time comparison from Experiment 2.**
2. **A completed notebook from Experiment 1.**